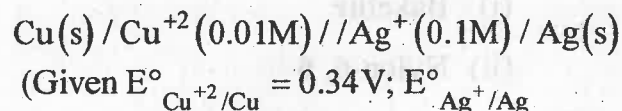


(4)

TCH-101

Weight of coal burnt = 1.28 gm, Weight of water taken = 350 gm, Weight of water equivalent of bomb and calorimeter = 2050 gm, Rise in temperature = 2.2°C, Fuse wire correction = 15 cal, Acid correction = 25 cal, H = 6.6% and Latent heat of condensation of steam = 580 cal/gm.

5. (a) Define and calculate the cell potential of the given cell at 25°C. Also write the half cell reactions of the following cell : (CO5)



- (b) What do you understand by Corrosion ? Explain it with the help of electro-chemical theory of corrosion. (CO5)

- (c). Write short notes on the following : (CO5)

(i) Concentration Cells

(ii) Fuel Cells

TCE-101

3180

H

Roll No.

TCH-101

**B. TECH. (FIRST SEMESTER)
END SEMESTER
EXAMINATION, Dec., 2023
ENGINEERING CHEMISTRY**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) On the basis of MOT diagram, explain why N_2 is diamagnetic in nature. Also find its bond order and magnetic nature.

(CO1)

P. T. O.

(2)

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- (b) What do you mean by H-bonding ?
Explain why, ortho-nitrophenol and para-nitrophenol can be able to separate through fractional distillation method.
(CO1)
- (c) Write short notes on the following : (CO1)
- (i) Band theory of metals
 - (ii) Applications of Spectroscopy
2. (a) Explain the zeolite process for softening of hard water and also discuss its advantages and disadvantages. (CO2)
- (b) Define and calculate the temporary and permanent hardness of a water sample which on analysis have the following :
(CO2)
- $$\text{Mg}(\text{HCO}_3)_2 = 9.2 \text{ ppm}$$
- $$\text{MgSO}_4 = 24.0 \text{ ppm}$$
- $$\text{CaSO}_4 = 3.40 \text{ ppm}$$
- $$\text{CaCl}_2 = 5.55 \text{ ppm}$$
- (c) Write short notes on the following : (CO2)
- (i) Boiler feed water
 - (ii) Calgon conditioning

(3)

TCH-101

3. (a) (i) What do you mean by Functionality of a monomer ? (CO3)
- (ii) Differentiate between thermoplastic and thermosetting polymers.
- (b) Write short notes on the following : (CO3)
- (i) Conducting polymers
 - (ii) Biodegradable polymers
- (c) Write the preparation, properties and uses of the following : (CO3)
- (i) Bakelite
 - (ii) Nylon-6, 6
4. (a) What do you mean by Biogas ? Explain the construction and working of the Biogas plant. (CO4)
- (b) Write short notes on the following : (CO4)
- (i) Biomass
 - (ii) CNG and LPG.
- (c) Define and calculate the HCV and LCV of a fuel, when tested in the laboratory for its calorific value in the bomb calorimeter, the following data were obtained : (CO4)